Planning as a Cognitive Cost that Discourages Goal Switching

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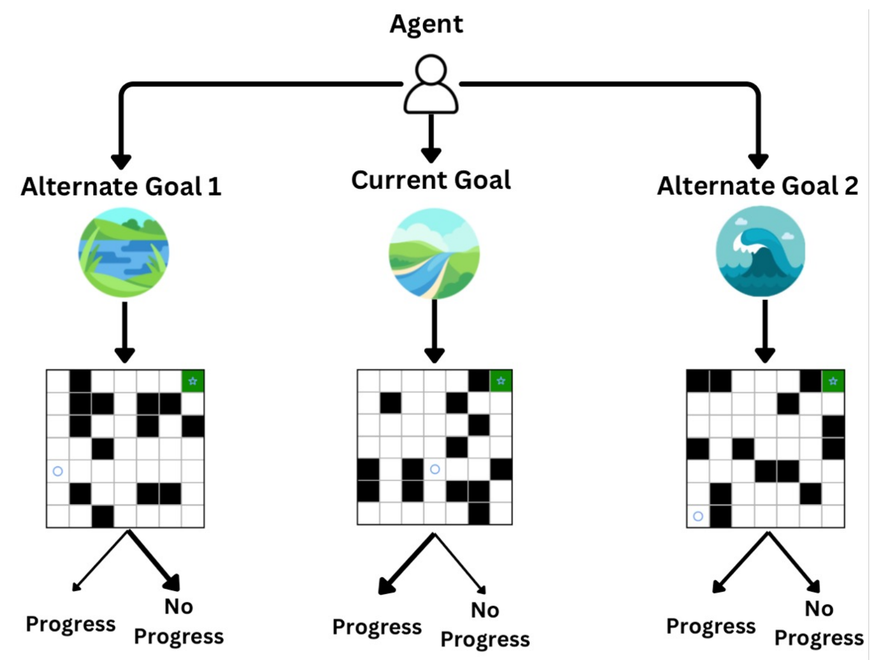
Introduction

- Goals are essential to guiding and sustaining human behavior[1]
- People show "stickiness" to active but suboptimal goals[1]
- Planning towards goals is cognitively costly, and people may be averse to these costs[2]

Does having to re-plan towards a new goal would exert a "planning cost" that discourages individuals from switching their goal?

Task

1. Choose to continue to abandon goal
2. Navigate GridWorld Maze
3. Receive Feedback



- **Non-planning Condition:** participants shown optimal route (plan) outlined in blue, and unable to deviate from it
- **Planning Condition:** participants not shown optimal route; have to create their own plans

Reinforcement Learning Models

- Rescorla-Wagner: $Q_t(g_i) = v_t(g_i) - c_t(g_i)$
- Momentum[3]: $Q_t(g_i) = m_t(g_i)v_t(g_i) - c_t(g_i)$
- Hybrid: $\big(1 - \omega\big(1 - m_t(g_i)\big)\big)\, v_t(g_i) - c_t(g_i)$
  $= \big(\omega \cdot m_t(g_i) + (1 - \omega)\big)v_t(g_i) - c_t(g_i)$

Where:

$$v_t(g_t) = v_{t-1}(g_t) + \alpha\big(r_t - v_{t-1}(g_t)\big)$$

$$m_t(g_i) = \frac{\text{Progress Made on Goal } i}{\text{Size of Goal } i}$$

$$c_t = \begin{cases} 0, & g_i = g_{t-1} \\ \kappa, & g_i \neq g_{t-1} \end{cases}$$

Note that these algorithms only model goal selection, not the actual mechanism of goal pursuit (i.e. maze navigation)

Results

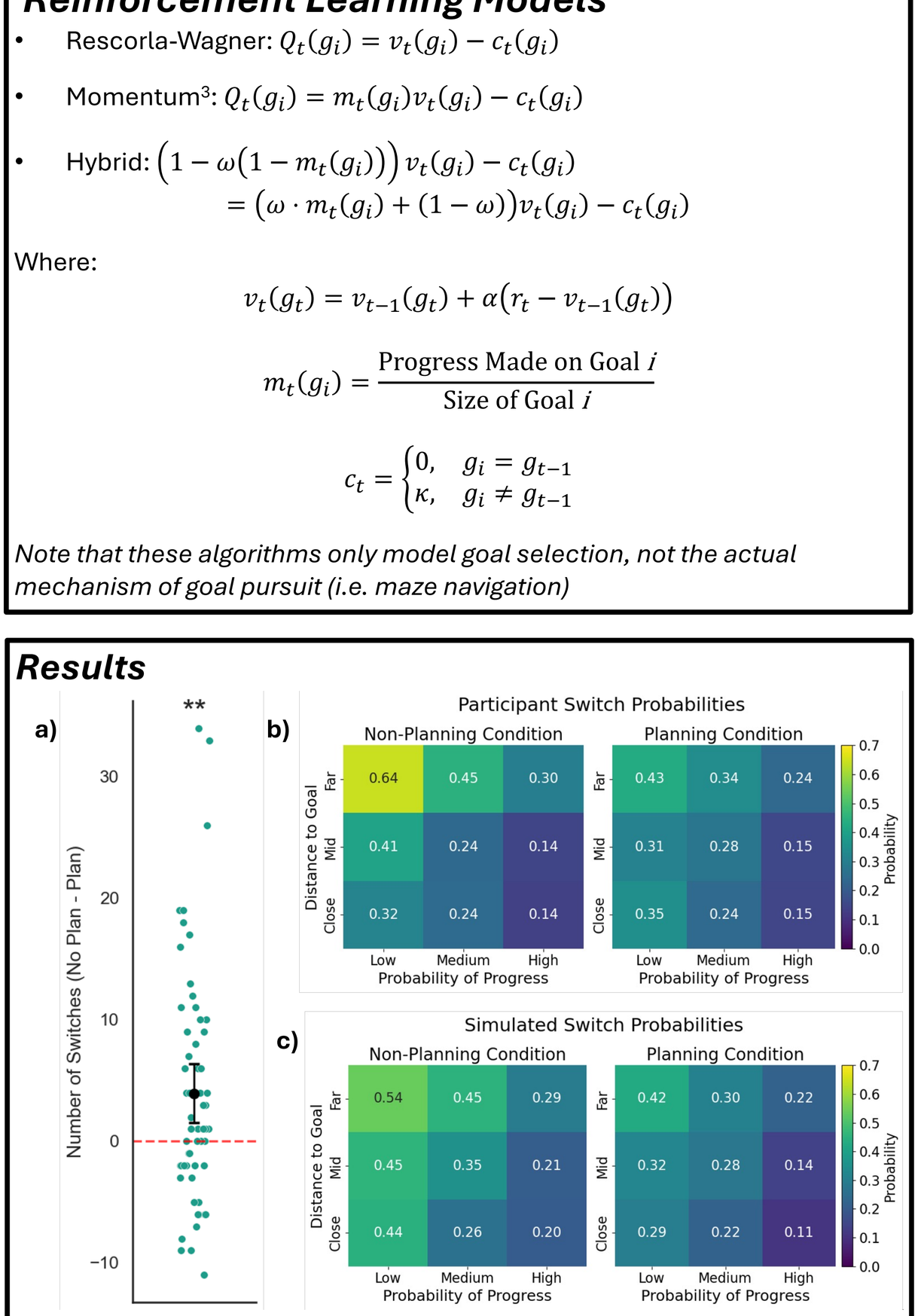


Figure 1a) People overall switch less frequently when they have to plan. b) Participants switch less often holding goal progress probability and accrued progress constant. c) Simulated goal-switch probabilities capture patterns in participant data

Results (cont.)

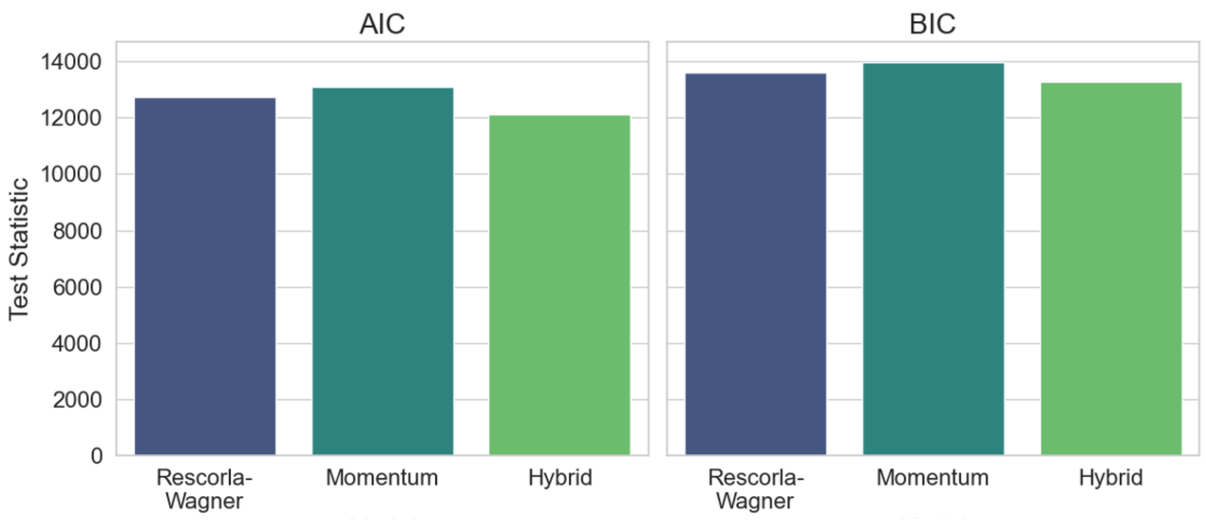


Figure 2) Left: AIC model comparison of the three reinforcement learning models (lower indicates better fit)
Right: BIC model comparison of the three reinforcement learning models.

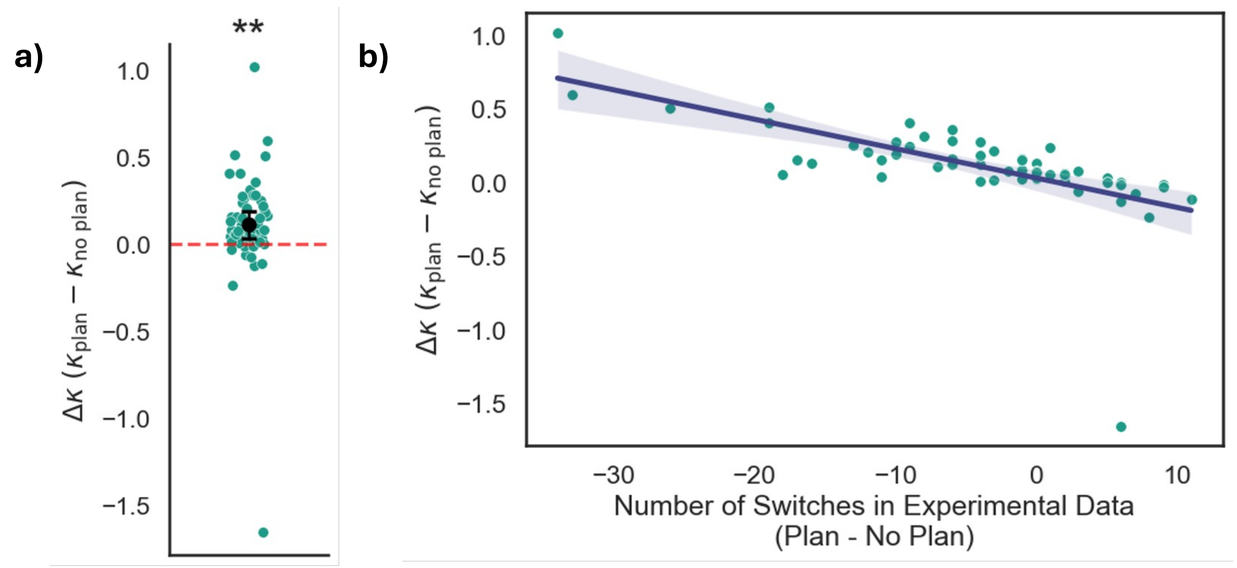


Figure 3a) Choice kernel values are higher when individuals have to plan.
b) Planning costs are negatively correlated with the frequency of switching

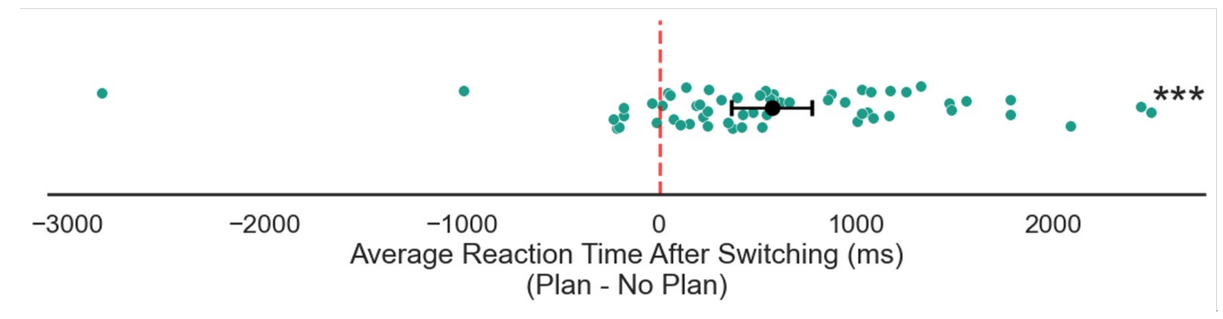


Figure 4) Reaction times after switching are on average greater when participants have to plan

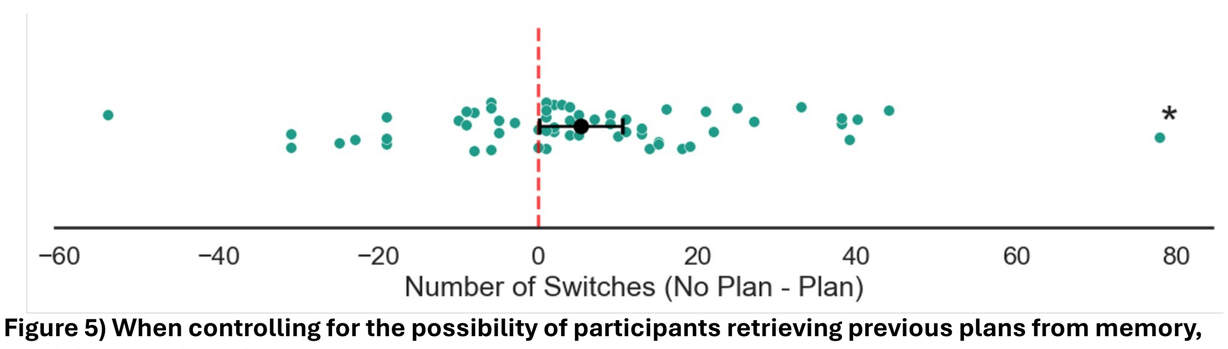


Figure 5) When controlling for the possibility of participants retrieving previous plans from memory, participants still switch less frequently when they plan

Discussion

- We conducted a behavioral experiment that demonstrated people are more likely to abandon their goal in situations where planning is not necessary (plans are provided for them), in contrast to situations in which they have to form their own plan
- Within-participant computational modeling captured goal persistence biases by treating planning costs as a value-based penalty, and demonstrated that this cost was higher when participants had to form their own plans
- Next step: Vary the complexity of these mazes to see if increased difficulty in creating plans will result in greater goal persistence and greater cost kernel values

Acknowledgments

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Citations

1. Holton, E., Grohn, J., Ward, H., Manohar, S. G., O'Reilly, J. X., & Kolling, N. (2024). Goal commitment is supported by vmPFC through selective attention.
2. Callaway, F., van Opheusden, B., Gul, S., Das, P., Krueger, P. M., Griffiths, T. L., & Lieder, F. (2022). Rational use of cognitive resources in human planning.
3. Aenugu, S., & O'Doherty, J. P. (2025). Building momentum: A computational account of persistence toward long-term goals.